

Scientific Life

Science from the bottom up

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How a scientific community is organized can affect whether disagreement drives progress or leads to isolation. Students and postdocs are uniquely placed to build communities that bridge theoretical and methodological divides. Drawing on a student-led society in consciousness science, we show how empowering early-career researchers can turn fragmentation into collaboration.

The social structure of scientific fragmentation

Scientific fields are often characterized by their discoveries, methods, and theories. Less visible, but equally important, are the social interactions and organizational structures of the researchers who produce them. Research in the science of science [1] has shown that the social structure of a scientific community shapes scientific outcomes: a fragmented field, in which researchers interact mostly with like-minded peers and rarely across community boundaries, limits scientific progress [2]. This pattern is partly self-reinforcing, as scientific networks tend to grow by densifying around existing expertise, which may constrain what gets studied next [1]. However, other patterns have the potential to counter this narrowing. For example, informal networks of advice and feedback reach further across a field

than coauthorship does, and are less hierarchical; and a researcher's position within them is closely tied to the impact of their work [3]. Therefore, scientific community structure can influence which ideas a researcher encounters, on whose work they build, and whether disagreements act as a catalyst for progress or settle into parallel research programs with little cross-communication [2].

Fragmentation is not merely an unintended by-product of a growing field; it is a structural equilibrium sustained by the organization and incentives of research communities, where the safer path of extending established traditions is favored over bridging across them [2,4]. This makes shared terminology and common standards harder to achieve, with different terms potentially describing the same construct, or different constructs possibly sharing a single term [5]. Consequently, conveying what is known becomes more difficult, not only within a field but also outside of it—to practitioners, policymakers, and the public [4]. The challenge, then, is not only to develop better theories or methods, but also to build the structures that make collaboration across diverse perspectives more likely [2,6].

The need for a bottom-up approach

Students and postdoctoral researchers are uniquely positioned to build communities that facilitate this kind of collaboration: their theoretical commitments are still forming, and the methods and perspectives they encounter have the potential to steer the field for decades to come. This is also the career stage at which scientists are most likely to produce work that transforms, rather than extends, established paradigms [7]. Therefore, empowering researchers at this stage to build communities across disciplines and frameworks has value at more than one level. For the researchers themselves, the ties they form beyond their immediate group can

predict their research impact [3]. For the field, collaborations that reach across such boundaries can produce more original work [6]. In addition, the perspectives, methods, and connections these researchers gain may in turn benefit their research groups and institutions.

However, this potential is constrained, as students and postdocs often encounter only a narrow part of their field at precisely the stage when breadth would matter most: incentives draw them to remain within the silos of their programs [4]; short-term positions and rapid publication timelines make sustained, integrative work risky to pursue [1,8]; and support for boundary-crossing collaborations is largely concentrated at later career stages [7]. This creates the need for bottom-up research communities that help students and postdocs (researchers in training) encounter unfamiliar perspectives and build cross-disciplinary collaborations.

Consciousness science as a case in point

Consciousness science can serve as a useful case study: over the past three decades, the field has matured rapidly, expanding across disciplines, paradigms, and theories [9]. Yet much of this progress has developed along parallel lines of research, with limited exchange between them [10]. At the same time, the field is increasingly asked to address high-stakes societal and policy questions—from clinical decision-making [11] to the possible sentience of artificial systems [12]. While both empirical and theoretical advances are essential, progress will also depend on whether the field's distinct lines of research can inform one another. Recent efforts have begun to address this fragmentation: adversarial collaborations, for example, bring competing theories into dialogue through shared experimental designs and jointly specified empirical tests [13]. However, initiatives such as these take effect not only through the

Diversity of MESEC Participants

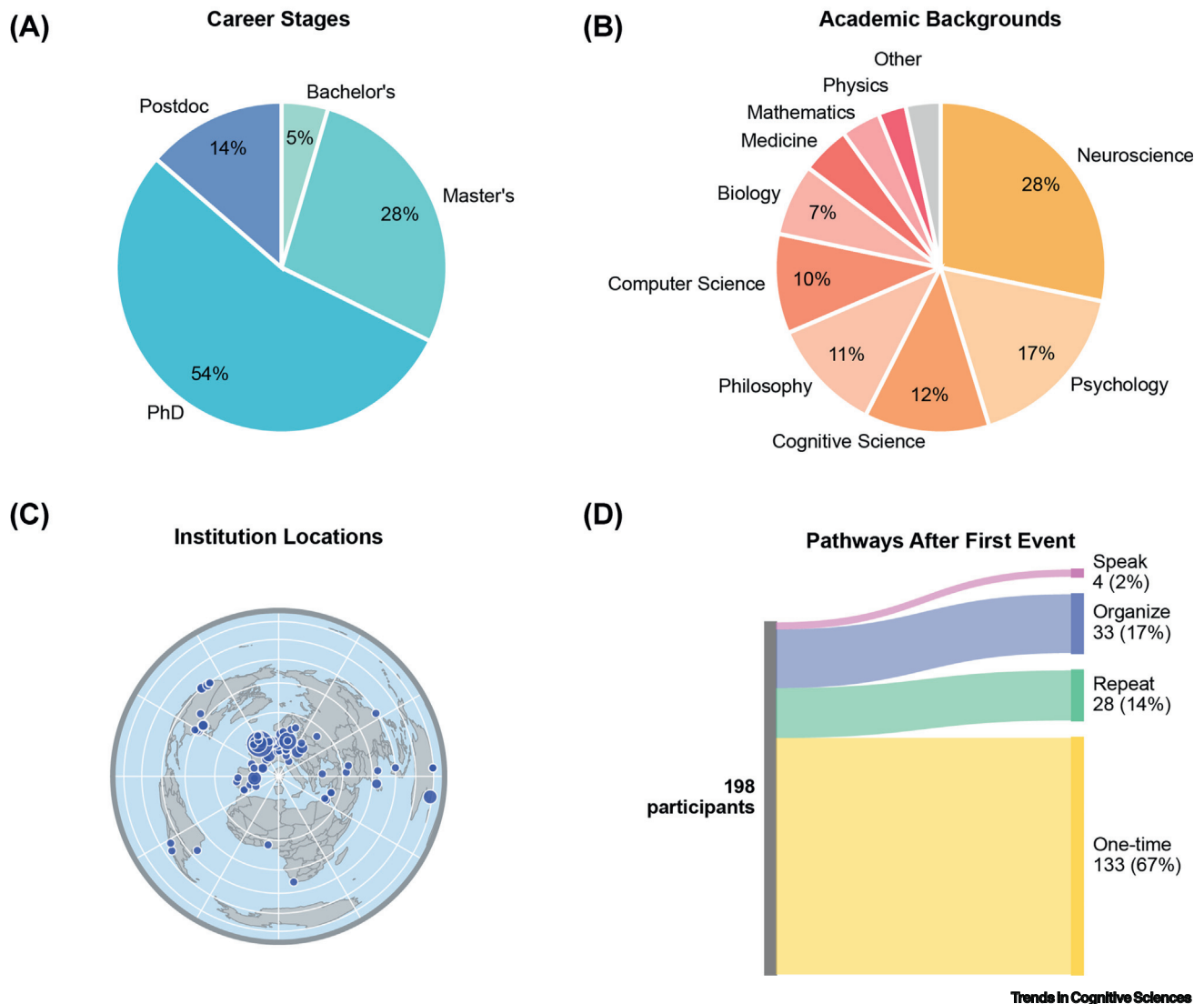


Figure 1. Diversity and engagement of MESEC participants. Data are from 2021 to 2026 (N = 198). (A) Career stages: bachelor's (5%), master's (28%), doctoral (54%), and postdoctoral (14%). (B) Academic backgrounds: participants represent a wide range of academic disciplines, reporting one to six fields ($M = 2.23$ and $SD = 1.03$). (C) Institution locations: participants represent 112 institutions across 30 countries. (D) Sustained community involvement: over one-third of participants remain active within the community, reconnecting at Residencies or Harbours (14%) and/or organizing following events (17%). As their careers progress, past participants return as speakers (2%). This ensures the sustainability of collaborations and MESEC as a whole. MESEC: Mediterranean Society for Consciousness Science.

efforts of those directly involved in them but also through how the rest of the field receives them. Reducing scientific fragmentation is a gradual, community-level

process that develops through ongoing contact [4], and venues for exchange of that kind remain scarce [2]. Bottom-up communities offer a complementary

approach: one that makes room for researchers themselves to build shared concepts and methods across differing frameworks together.

From principle to practice: MESEC

Despite the challenges, bottom-up initiatives to create such early-career communities are possible. In consciousness science, one example is the Mediterranean Society for Consciousness Science (MESEC). Founded in 2021, MESEC was our attempt to build what we were missing: exchange and collaboration that were broader than any single institution or theoretical tradition could offer on its own. Since its first Workshop, MESEC has grown into a global community of nearly 300 researchers; more than two-thirds of them are researchers in training who span over 100 institutions across 30 countries and a wide range of disciplines (Figure 1).

The community now runs three annual series of events, each designed to support a different mode of scientific exchange: Winter Schools introduce participants to the methodological and theoretical breadth of the field; Workshops are dedicated to making progress on frontier questions; and Residencies connect participants across cohorts to facilitate long-term collaboration. Each event brings together around 20 students and postdocs, and a handful of invited senior researchers, who spend a full week living and working together in a secluded setting away from their home institutions. This extended, intensive format creates a space for exchanges that brief conferences do not typically allow: researchers new to the field build lasting relationships; researchers from different traditions develop a shared vocabulary and have time to find common ground; and disagreements can turn into productive research questions. A bursary program lowers financial barriers to participation and helps widen the community across disciplines, career stages, geographies, and demographic backgrounds. In addition to MESEC's internal events, Harbours extend support to researchers in training at major conferences: the community

reserves shared accommodation, easing the cost and logistics of attending important events in the field, and turning the stay itself into an opportunity for connection.

The organizational structure of MESEC embodies the same bottom-up approach: leadership is held by students and postdocs, and events are conceived and run by rotating teams of past participants. These teams are provided with support to secure event-specific grants in their own names, giving researchers in training hands-on experience with grant writing, budget management, and project leadership—skills that institutional structures typically reserve for more senior investigators [7]. The continuity of the community is therefore carried by shared values rather than fixed roles. Rotation of this kind keeps the community close to emerging questions and methods, while giving researchers in training the opportunity to influence the field they are becoming part of.

The effects of this approach extend beyond the events themselves. Collaborations initiated through MESEC have produced a growing body of publications spanning different theoretical and methodological perspectives (e.g., [14–15]); a recent Workshop served as the basis for a special issue bridging theoretical work and clinical application in the journal *Philosophy and the Mind Sciences*; and former participants established complementary early-career regional initiatives, creating additional accessible spaces for sustained exchange. Importantly, the effectiveness of this collaborative model is not specific to consciousness science. Its underlying principles—bringing different perspectives into extended dialogue, supporting connections beyond any single event, and shared, bottom-up ownership—can be applied in any scientific field, wherever researchers take on the work of building such communities.

Concluding remarks

Researchers in training are an epistemic force in scientific fields. The questions they ask, the collaborations they form, and the intellectual habits they develop will shape how science is done in the future. We believe that it is time to invest in them, and in the science they can practice together.

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Declaration of interests

The authors declare no competing interests.

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